Year 12 Maths - Quadratics

Similar questions - Discriminant and the number of real roots

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The equation $x^2 - 2(k-1)x + (k+5) = 0$, where k is a constant, has two distinct real roots.

(a) Show that $k^2 - 3k - 4 > 0$. **[3]**

(b) Hence find the set of possible values of k. **[2]**

(c) Find the repeated root of the equation when $k = 4$. **[2]**

2 Consider the equation $x^2 - (k+3)x + 2(k+1) = 0$.

(a) Show that the discriminant of the equation is $(k-1)^2$. **[3]**

(b) Hence explain why the equation has real roots for every value of k. **[1]**

(c) Find the value of k for which the roots are equal, and solve the equation for this value of k. **[2]**

Question 1

(a) $b^2 - 4ac = 4(k-1)^2 - 4(k+5) = 4k^2 - 12k - 16 = 4(k^2 - 3k - 4)$. Two distinct real roots requires $b^2 - 4ac > 0$, so $k^2 - 3k - 4 > 0$

(b) $(k-4)(k+1) > 0$, so $k < -1$ or $k > 4$

(c) When $k = 4$: $x^2 - 6x + 9 = 0$, so $(x-3)^2 = 0$ and $x = 3$

Question 2

(a) $b^2 - 4ac = (k+3)^2 - 8(k+1) = k^2 + 6k + 9 - 8k - 8 = k^2 - 2k + 1 = (k-1)^2$

(b) A square is never negative, so $b^2 - 4ac \geq 0$ for all k

(c) Equal roots when $(k-1)^2 = 0$, so $k = 1$. Then $x^2 - 4x + 4 = 0$, so $x = 2$